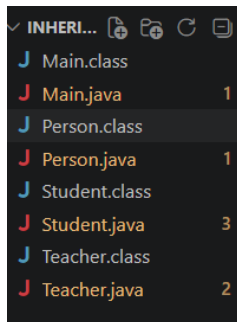


Nama : Siti Nurmeilani
NIM : 20240040243
Kelas : TI24D
Mata Kuliah : PBO 5

Program dibuat menggunakan konsep OOP Inherentance.
Class Person menjadi parent class, sedangkan Student dan Teacher menjadi child class.

Source Code



1. Person.java

```
Person.java > Java > Person
1  public class Person {
2
3      protected String name;
4      protected String address;
5
6      // Constructor
7      public Person(String name, String address) {
8          this.name = name;
9          this.address = address;
10     }
11
12     // Getter name
13     public String getName() {
14         return name;
15     }
16
17     // Getter address
18     public String getAddress() {
19         return address;
20     }
21
22     // Setter address
23     public void setAddress(String address) {
24         this.address = address;
25     }
26
27     // toString
28     public String toString() {
29         return name + "(" + address + ")";
30     }
31 }
32
```

2. Student.java

```
1 Student.java > Language Support for Java(TM) by Red Hat > Student
2 public class Student extends Person {
3
4     private int numCourses = 0;
5     private String[] courses = new String[10];
6     private int[] grades = new int[10];
7
8     // Constructor
9     public Student(String name, String address) {
10         super(name, address);
11     }
12
13     // Tambah course dan nilai
14     public void addCourseGrade(String course, int grade) {
15         courses[numCourses] = course;
16         grades[numCourses] = grade;
17         numCourses++;
18     }
19
20     // Tampilkan nilai
21     public void printGrades() {
22         for (int i = 0; i < numCourses; i++) {
23             System.out.println(courses[i] + " : " + grades[i]);
24         }
25     }
26
27     // Hitung rata-rata
28     public double getAverageGrade() {
29
30         int sum = 0;
31
32         for (int i = 0; i < numCourses; i++) {
33             sum += grades[i];
34         }
35
36         return (double) sum / numCourses;
37     }
38
39     // toString
40     public String toString() {
41         return "Student: " + super.toString();
42     }
43 }
```

3. Teacher.java

```
1 public class Teacher extends Person {
2     private int numCourses = 0;
3     private String[] courses = new String[10];
4
5     // Constructor
6     public Teacher(String name, String address) {
7         super(name, address);
8     }
9
10    // Tambah mata kuliah
11    public boolean addCourse(String course) {
12
13        for (int i = 0; i < numCourses; i++) {
14            if (courses[i].equals(course)) {
15                return false;
16            }
17        }
18
19        courses[numCourses] = course;
20        numCourses++;
21
22        return true;
23    }
24
25    // Hapus mata kuliah
26    public boolean removeCourse(String course) {
27
28        int index = -1;
29
30        for (int i = 0; i < numCourses; i++) {
31            if (courses[i].equals(course)) {
32                index = i;
33                break;
34            }
35        }
36
37        if (index == -1) {
38            return false;
39        }
40
41        for (int i = index; i < numCourses - 1; i++) {
42            courses[i] = courses[i + 1];
43        }
44
45        numCourses--;
46
47        return true;
48    }
49
50    // toString
51    public String toString() {
52        return "Teacher: " + super.toString();
53    }
54 }
```

4. Main.java

```
1 public class Main {
2     public static void main(String[] args) {
3
4         Scanner input = new Scanner(System.in);
5
6         // ===== MAHASISWA =====
7
8         System.out.println(x: "=== INPUT DATA MAHASISWA ===");
9
10        System.out.print(s: "Nama Mahasiswa : ");
11        String namaMhs = input.nextLine();
12
13        System.out.print(s: "Alamat Mahasiswa : ");
14        String alamatMhs = input.nextLine();
15
16        Student s1 = new Student(namaMhs, alamatMhs);
17
18        System.out.print(s: "Jumlah Mata Kuliah : ");
19        int jumlahMK = input.nextInt();
20        input.nextLine();
21
22        for (int i = 0; i < jumlahMK; i++) {
23
24            System.out.print(s: "Nama Mata Kuliah : ");
25            String mk = input.nextLine();
26
27            System.out.print(s: "Nilai : ");
28            int nilai = input.nextInt();
29            input.nextLine();
30
31            s1.addCourseGrade(mk, nilai);
32        }
33
34        System.out.println();
35        System.out.println(s1);
36
37        System.out.println(x: "Daftar Nilai:");
38        s1.printGrades();
39
40        System.out.println("Rata-rata Nilai : " + s1.getAverageGrade());
41
42        // ===== DOSEN =====
43
44        System.out.println(x: "\n=== INPUT DATA DOSEN ===");
45
46        System.out.print(s: "Nama Dosen : ");
47        String namaDosen = input.nextLine();
48
49        System.out.print(s: "Alamat Dosen : ");
50        String alamatDosen = input.nextLine();
51
52        Teacher t1 = new Teacher(namaDosen, alamatDosen);
53
54        System.out.print(s: "Jumlah Mata Kuliah yang Diajar : ");
55        int jumlahAjar = input.nextInt();
56        input.nextLine();
57
58        for (int i = 0; i < jumlahAjar; i++) {
59
60            System.out.print(s: "Nama Mata Kuliah : ");
61            String mk = input.nextLine();
62
63            boolean hasil = t1.addCourse(mk);
64
65            if (hasil) {
66                System.out.println(x: "Berhasil ditambahkan");
67            }
68        }
69    }
70 }
```

Compile :

```
PS C:\Users\ASUS VIVOBOOK\OneDrive\ドキュメント\Inheritance5> javac Main.java
PS C:\Users\ASUS VIVOBOOK\OneDrive\ドキュメント\Inheritance5> java Main
=== INPUT DATA MAHASISWA ===
Nama Mahasiswa : Lani
Alamat Mahasiswa : Sukabumi
Jumlah Mata Kuliah : 2
Nama Mata Kuliah : PBO
Nilai : 90
Nama Mata Kuliah : Big Data
Nilai : 80

Student: Lani(Sukabumi)
Daftar Nilai:
PBO : 90
Big Data : 80
Rata-rata Nilai : 85.0

=== INPUT DATA DOSEN ===
Nama Dosen : Yulhan
Alamat Dosen : Sukabumi
Jumlah Mata Kuliah yang Diajar : 2
Nama Mata Kuliah : PBO
Berhasil ditambahkan
Nama Mata Kuliah : Big Data
Berhasil ditambahkan

Teacher: Yulhan(Sukabumi)
PS C:\Users\ASUS VIVOBOOK\OneDrive\ドキュメント\Inheritance5>
```

Seluruh fungsi pada diagram UML telah berhasil dibuat dan dijalankan seperti constructor, getter, setter, addCourseGrade(), printgGrades(), getAverageGrade(), addCourse(), removeCourse(), dan toString().

Program berhasil dibuat dan dijalankan menggunakan konsep inheritance pada Java.